RESCUE ROVER – PROJECT REPORT (MEDIUM LENGTH)

1. ABSTRACT

The Rescue Rover is a compact robotic vehicle designed for flood rescue, landslide search, canal inspection, and drainage cleaning. It can safely enter hazardous and narrow areas where humans cannot go. Using sensors, a camera, wireless communication, and lighting, it provides real-time information to rescue teams. A Flutter-based mobile app was developed for live video streaming and remote control. The rover's functionality is programmed using Arduino/C language on an ESP32 microcontroller.

2. INTRODUCTION

Natural disasters such as floods and landslides create areas too dangerous for human entry. Narrow canals and sewage lines pose similar risks. The Rescue Rover provides a safe, low-cost solution for inspecting these areas using sensors, cameras, and wireless communication.

3. PROBLEM STATEMENT

Human entry into flooded, landslide-affected, or narrow areas is unsafe.

Manual inspection is slow and risky.

A remotely controlled robot can enter inaccessible areas and relay live data to rescue teams.

4. OBJECTIVES

Design a robot capable of entering narrow and dangerous environments.

Provide live video feed for rescue and inspection.

Detect humans using a thermal sensor.

Avoid obstacles with an ultrasonic sensor.

Detect harmful gases.

Support flood, landslide, canal, and drainage operations.

Enable remote control via a Flutter app.

5. PROPOSED SYSTEM

ESP32 microcontroller is the main controller.

Rover equipped with motors, sensors, FPV camera, and LED lights.

Flutter mobile app for live streaming and movement control.

Arduino/C code handles motor control, sensor readings, RF communication, and video streaming.

6. BLOCK DIAGRAM (TEXT FORM)

Battery → ESP32 Controller → Motor Driver → Motors

ESP32 → Ultrasonic Sensor

ESP32 → Thermal Sensor

ESP32 → FPV Camera + LED Lights

ESP32 ↔ NRF24L01 RF Module → Remote Controller / Flutter App

7. MAJOR COMPONENTS

ESP32 Microcontroller – main controller

NRF24L01 RF Module – long-range communication

Motor Driver (L298N) – controls motors

DC Motors – movement

Ultrasonic Sensor – obstacle detection

Thermal Imaging Sensor – human detection

FPV Camera – real-time video feed

LED Lights – illumination

Gas Sensor – harmful gas detection

Battery Pack – powers the rover

8. WORKING PRINCIPLE

Rover powered by a rechargeable battery pack.

ESP32 initializes all sensors and motors.

Remote commands sent via NRF24L01 or WiFi from the Flutter app.

Motor driver controls movement: forward, backward, left, right.

Ultrasonic sensor avoids obstacles.

FPV camera streams live video to the Flutter app.

LED lights enhance visibility.

Thermal sensor detects humans or animals.

Gas sensor detects hazardous gases.

9. MOBILE APP & ARDUINO INTEGRATION

Flutter App:

• Controls rover movements

• Displays live video

• Shows sensor readings

• Sends alerts for dangerous gas detection

Arduino/C Code:

• Controls motors and sensors

• Handles RF communication

• Streams camera feed to the app

Integration:

App sends commands → Rover executes → Data/video returns to app

Allows safe, real-time rescue operations

10. APPLICATIONS

• Flood rescue operations

• Landslide search and victim detection

• Canal and drainage inspection

• Sewage system monitoring

• Disaster management support

• Building collapse inspection

11. ADVANTAGES

• Safe operation in hazardous areas

• Reduces risk for rescue teams

• Real-time visual and sensor data

• Works day and night

• Low-cost, modular, and easy to upgrade

12. FUTURE SCOPE

• Add robotic arm for object removal

• Autonomous navigation using AI

• Underwater or waterproof versions

• GPS mapping for outdoor rescue

• Enhanced thermal and night vision sensors

13. CONCLUSION

The Rescue Rover provides a safe, efficient, and practical solution for rescue and inspection tasks. With Arduino-coded control, sensors, camera, LED lights, and Flutter app integration, it can operate in areas inaccessible to humans, assisting rescue teams during floods, landslides, and blocked canals.